Lecture 1

- book: W.D Callister, Materials Science & Engineering An Introduction 5th to 9th ed, J. W
- In tutorial Problems: - based on Assignments
  - every 2nd tutorial (first is Sep 21)
- Midterm: Friday Oct 30th
- term assignment: Nov 27th

- What is materials?

Oil, gases, pharmaceutitacls: matter

Iron, copper, polymer, cement: material

materials - can be put into certain geometric shapes
- the product has some function

Property: Response of a material to an external effect, such as

- mechanical
- electrical
- optical
- thermal
- magnetic

- independent of material's shape & size

Why study Mat science: - Important to understand capabilities and limitations of materials

- An understanding of Materials science helps us to design better components, parts, devices, ect.
  - how can we make smt stronger / lighter?
  - how do elements come tg to form alloys?
  - why do sum mats have vastly different properties than others?